What GreenLCC India is about

A decision tool that finds the cheapest way, over a building's whole life, to reach an IGBC green rating, and measures how safe that answer is.

The problem

When owners choose solar panels, air-conditioning, glazing or lighting, they usually pick by lowest purchase price. Over a 25-year life, though, electricity, maintenance and replacements cost far more than the purchase itself, so the cheapest to buy is often the most expensive to own. Prices also change: electricity tariffs rise, interest rates move, and equipment wears out at different ages. And owners often want a green rating (IGBC Certified, Silver, Gold or Platinum) on top, which means choosing a whole combination of measures, not just one.

What the tool does

GreenLCC takes a building, its energy and water prices and a list of green options, and answers four questions:

- **What is the cheapest way to reach my target IGBC level?** It checks every legal combination, over 1,000 in the demo, so the answer is exact, not a guess.
- **What does choosing by price alone cost me?** It places the cheapest-upfront route beside the lowest-life-cycle-cost route.
- **Is the extra upfront money worth it?** It works out what the extra money earns each year and compares it with the owner's required return (the MARR).
- **How safe is the answer?** It shows where the ranking would flip and tests thousands of random futures.

The demo result

For a 12,000 m^2 Pune office aiming for IGBC Gold (all prices are sourced estimates):

	Cheapest upfront	Recommended (lowest life-cycle cost)
Upfront cost	₹1.81 Cr	₹4.57 Cr
Life-cycle cost (present worth, 25 years, 10% MARR)	−₹2.97 Cr	−₹6.66 Cr
IGBC result	Gold, 70 points	Gold, 76 points
What the extra ₹2.75 Cr earns	-	13.8% a year (MIRR) vs the 10% required
Flips only if the required return rises above	-	22.3%
Wins in simulated futures	-	100% of 3,000

A negative life-cycle cost means savings exceed all costs: the building ends up better off. The recommended mix is the water-cooled chiller, LED lighting with sensors, energy metering and BMS, cool roof, 200 kWp rooftop solar and low-flow fixtures.

Who it is for

User	How they use it
Building owners and developers	Choose between design options and see the whole-life consequence before committing.
Architects and MEP consultants	Justify a specification to a client with numbers, not opinion.
Procurement teams	Compare bids on life-cycle cost instead of price alone.
Sustainability consultants	Plan the cheapest route to an IGBC rating.
Facility managers	Plan replacement and operating budgets.

How it connects to the course (CE 725)

Course idea	Where it shows up in the tool
Time value of money, nominal vs real rates	Everything is discounted at the MARR; the real MARR is shown using the Fisher equation
Present worth, planning horizon, salvage	Life-cycle cost over one study period (25 years) with salvage and re-purchases
Equivalent annual worth	"Cost per year" (EUAC) beside every present worth
MARR, IRR, incremental IRR, MIRR	The headline: the return on the extra upfront spend versus the MARR
Benefit-cost ratio, discounted payback	Shown for each measure on Compare measures
Sensitivity, break-even, Monte Carlo	The "Can we trust it?" page

How AI was used

AI helped the team research the problem, write and debug the code and draft the documents. Every number in the tool comes from fixed formulas, not from AI, and each is covered by automatic tests, including worked examples from the course notes. Inside the running tool, AI is optional and only polishes the wording of the owner memo; it is told never to change a number, and without an API key a built-in template is used. The team logs significant AI prompts and whether each output was accepted, modified or rejected.

Honest limits

- Prices, tariffs and equipment costs are sourced estimates for a demo building; replace them with real project data.
- The IGBC point tables are illustrative, not the official reference guide.
- The view is the owner's, before tax: depreciation, tax and financing are outside the model.
- Savings of different measures are added up; interaction between measures is not modelled.